Working principle

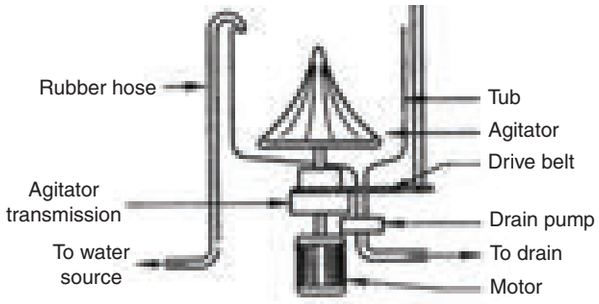


Fig. 4.22

In this, clothes to be washed along with soap powder are put in the inner container of the washing machine. Dirty clothes are rotated up and down with soap powder by water. This process takes place for a fixed time duration.

The inner tub of the washing machine has many holes. Water is sucked out to the outer tub from the inner container by centrifugal force. Water is drained out from the outer tub through a drain vent pipe. When the dirty water is drained out, fresh water is filled into the inner tub again. Agitator rinses the clothes. In this process the chemicals present in the soap are removed. After this process is completed, water in the machine is again drained out through the outer pipe.

After water is drained, the clothes kept in the container are rotated at a high speed and the water in the clothes is removed by the centrifugal force. After this cycle takes place for a certain period of time, the excess water in the clothes are squeezed out with the help of a time controlling device. Now the clothes are washed and are ready to be dried.

2. Front loading washing machine



Fig. 4.22 **Front loading washing machine**

a) Construction

The structure and functioning of front loading washing machine are similar to that of top loading washing machine.

A cylindrical type of vessel is used in the front loading washing machine instead of the agitator in the upper loading washing machine. With the help of a roller-shaped container, the cylinder spins. As this event continues, the clothes are well washed in soapy water.

The impeller is fixed in the cylindrical vessel inside a front loading washing machine. This cylinder vessel is in horizontal position. The impeller fixed to this rotating vessel mixes detergent with water and rubs the clothes together to remove the dirt.

In some types of front loading washing machine, a heating element is placed on the bottom of the pipe to get warm water. Washing the clothes with warm water cleans the clothes quickly.

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